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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Winter Examination – 2022</b><br><b>Course: B. Tech.      Branch : Information Technology      Semester :VI</b><br><b>Subject Code &amp; Name: BTITOE604A Compiler Design</b><br><b>Max Marks: 60      Date:19/7/2023      Duration: 3 Hr.</b>                                                                |                                                                                                                                                                                                   |            |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                                                                                   |            |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                   | (Level/CO) | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                                                                            |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Draw a block diagram of phases of compiler and indicate main function of each phase.                                                                                                              | CO1        | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Define compiler. List some commonly used compiler construction tools.                                                                                                                             | CO1        | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Differentiate between compiler and interpreter                                                                                                                                                    | CO3        | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                   |            |           |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                                                                            |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain in brief input buffering mechanism in lexical analyser.                                                                                                                                   | CO1        | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | What is the role of regular expression in lexical analysis? Explain with examples.                                                                                                                | CO2        | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Illustrate transition diagram with the help of example and draw the transition diagram for the regular definition of Relop : $<   \leq   >   \geq   ==$                                           | CO2        | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                   |            |           |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                                                                            |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | What is left factored grammar? Explain steps to get a left factored grammar with the help of example.                                                                                             | CO2        | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Differentiate top-down parser and bottom-up parser                                                                                                                                                | CO3        | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | What is left recursive grammar. Eliminate left recursion and Find FIRST and FOLLOW for the following grammar.<br>$E \rightarrow E+T/T$<br>$T \rightarrow T * F / F$<br>$F \rightarrow id / ( E )$ | CO3        | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                   |            |           |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                                                                            |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain in brief about Synthesized and Inherited Attributes.                                                                                                                                      | CO1        | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | What is SDT? Explain in brief L-attributed grammar                                                                                                                                                | CO2        | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Write the three address code sequence for the statement $x = y * z + y * -z$ . Also                                                                                                               | CO3        | 6         |

|             |                                                                                                                |            |           |
|-------------|----------------------------------------------------------------------------------------------------------------|------------|-----------|
|             | give its quadruple and triple representation.                                                                  |            |           |
|             |                                                                                                                |            |           |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                         |            | <b>12</b> |
| <b>A)</b>   | What is the purpose of code optimization? Explain the DAG representation of basic blocks with example.         | <b>CO1</b> | <b>6</b>  |
| <b>B)</b>   | Explain the following terms:<br>(i) Common sub expression (ii) dead code elimination                           | <b>CO2</b> | <b>6</b>  |
| <b>C)</b>   | What is role of flow graph in the basic block? Explain how a given program can be converted into a Flow Graph? | <b>CO3</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                             |            |           |

|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               |                |           |
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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Winter Examination – 2023</b><br><b>Course: B. Tech.      Branch : Information Technology      Semester : VI</b><br><b>Subject Code &amp; Name: BTITOE604A Compiler Design</b><br><b>Max Marks: 60      Date:23/1/2024      Duration: 3 Hr.</b>                                                 |                                                                               |                |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                               |                |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               | (Level/<br>CO) | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                        |                | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain language translation system in detail.                                | <b>CO1</b>     | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Explain the terms: Lexeme, Token, Pattern, Symbol table, Syntax, Backpatching | <b>CO1</b>     | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Write a note on LEX.                                                          | <b>CO3</b>     | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               |                |           |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                        |                | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain with example when we need to rewrite the grammar and why?             | <b>CO1</b>     | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Explain the importance of CFG in compilation with examples.                   | <b>CO2</b>     | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Explain what is handle and handle pruning using operator precedence parsing.  | <b>CO2</b>     | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               |                |           |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                        |                | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain the Bottom-Up evaluation of inherited attributes.                     | <b>CO2</b>     | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Explain Top-Down translation in detail.                                       | <b>CO3</b>     | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | What is SDD. Explain with example.                                            | <b>CO3</b>     | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               |                |           |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                        |                | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain various parameter passing techniques with example.                    | <b>CO1</b>     | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Write a short note on intermediate code generation phase of compiler.         | <b>CO2</b>     | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Write a note on the simple type checker.                                      | <b>CO3</b>     | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                               |                |           |
| <b>Q. 5</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                        |                | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Explain issues in the design of code generation.                              | <b>CO1</b>     | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | Write a short note on principal sources of optimization.                      | <b>CO2</b>     | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Differentiate between compiler and interpreter                                | <b>CO3</b>     | <b>6</b>  |
| <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                                                 |                                                                               |                |           |

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular & Supplementary Summer 2024**

**Course:**B. Tech                      **Branch :** Information Technology                      **Semester :** VI

**Subject Code & Name:** BTITOE604A    **Compiler Design**

**Max Marks:** 60

**Date:** 21/06/2024

**Duration:** 03 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

(CO)                      Marks

**Q. 1 Solve Any Two of the following.**

**12**

- A) Show the phases of compiler with the help of following example.

CO1                      06

position=initial +rate\*60

- B) Explain Tokens, Patterns, and Lexemes with an example.

CO2                      06

- C) What is Input Buffering? Explain Buffer pairs with a diagram.

CO2                      06

**Q.2 Solve Any Two of the following.**

**12**

- A) Explain the Role of the Parser with neat diagram.

CO2                      06

- B) What is Left factoring and solve the following

CO3                      06

$S \rightarrow aSSbS \mid aSaSb \mid abb \mid b$

- C) Consider the following parsing table, Parse the string  $a^*(b+c)$  where the terminal a, b, c are considered as id

CO2                      06

| State | Action |    |    |    |     |     | GoTO |   |    |
|-------|--------|----|----|----|-----|-----|------|---|----|
|       | id     | +  | *  | (  | )   | \$  | E    | T | F  |
| 0     | s5     |    |    | s4 |     |     | 1    | 2 | 3  |
| 1     |        | s6 |    |    |     | acc |      |   |    |
| 2     |        | r2 | s7 |    | r2  | r2  |      |   |    |
| 3     |        | r4 | r4 |    | r4  | r4  |      |   |    |
| 4     | s5     |    |    | s4 |     |     | 8    | 2 | 3  |
| 5     |        | r6 | r6 |    | r6  | r6  |      |   |    |
| 6     | s5     |    |    | s4 |     |     |      | 9 | 3  |
| 7     | s5     |    |    | s4 |     |     |      |   | 10 |
| 8     |        | s6 |    |    | s11 |     |      |   |    |
| 9     |        | r1 | s7 |    | r1  | r1  |      |   |    |
| 10    |        | r3 | r3 |    | r3  | r3  |      |   |    |
| 11    |        | r5 | r5 |    | r5  | r5  |      |   |    |

(1)  $E \rightarrow E + T$

(4)  $T \rightarrow F$

(2)  $E \rightarrow T$

(5)  $F \rightarrow (E)$

(3)  $T \rightarrow T * F$

(6)  $F \rightarrow id$

Where E is start symbol.

|                                                                                                             |            |           |
|-------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 3 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Solve the following SDT and draw the annotated parse tree with dependency flow direction.                | <b>CO2</b> | <b>06</b> |
| $(3+5)*4*(2*(3+5))*(6+4)$                                                                                   |            |           |
| <b>Note: Use the grammar from Q.2.C</b>                                                                     |            |           |
| B) Explain Synthesized, Inherited attribute with Simple Desk Calculator example.                            | <b>CO2</b> | <b>06</b> |
| C) What Dependency Graph explain it with an example.                                                        | <b>CO3</b> | <b>06</b> |
| <b>Q.4 Solve Any Two of the following.</b>                                                                  |            | <b>12</b> |
| A) Construct DAG, Three address code, Quadruples, Triples, and Indirect triples of the following statement: | <b>CO3</b> | <b>06</b> |
| $a+a*(b-c)+(b-c)*d$                                                                                         |            |           |
| B) Explain Types and Declaration.                                                                           | <b>CO1</b> | <b>06</b> |
| C) Write a short note on Control Flow.                                                                      | <b>CO1</b> | <b>06</b> |
| <b>Q. 5 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Explain issues in the design of code generation.                                                         | <b>CO3</b> | <b>06</b> |
| B) Explain Principal sources of optimization.                                                               | <b>CO2</b> | <b>06</b> |
| C) Write a short note on Peephole optimization.                                                             | <b>CO3</b> | <b>06</b> |

\*\*\*\*\* End \*\*\*\*\*

Course: B. Tech.

Branch: Information Technology

Semester: VI

Subject Code &amp; Name: BTITOE604A Compiler Design

Max Marks: 60

Date: 26/12/2024

Duration: 3 Hr.

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                | (Level /CO)          | Marks     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                         | <b>CO1, CO2, CO3</b> | <b>12</b> |
| <b>1</b>    | <b>Which phase of the compiler is responsible for checking the token in a program?</b>                                         |                      | <b>1</b>  |
|             | a. Lexical Analysis      b. Syntax Analysis      c. Semantic Analysis      d. Code Generation                                  |                      |           |
| <b>2</b>    | <b>What does a syntax analyzer do?</b>                                                                                         |                      | <b>1</b>  |
|             | a. Generates target code      b. Checks for syntax errors      c. Breaks input into tokens      d. Generates intermediate code |                      |           |
| <b>3</b>    | <b>Which of the following tools is used to generate a syntax analyzer?</b>                                                     |                      | <b>1</b>  |
|             | a. YACC      b. LEX      c. GDB      d. GCC                                                                                    |                      |           |
| <b>4</b>    | <b>What is the output of a lexical analyzer?</b>                                                                               |                      | <b>1</b>  |
|             | a. Parse Tree      b. Tokens      c. Assembly Code      d. Intermediate Code                                                   |                      |           |
| <b>5</b>    | <b>Which phase is responsible for type checking and scope resolution?</b>                                                      |                      | <b>1</b>  |
|             | a. Syntax Analysis      b. Semantic Analysis      c. Lexical Analysis      d. Code Optimization                                |                      |           |
| <b>6</b>    | <b>What is an intermediate code representation?</b>                                                                            |                      | <b>1</b>  |
|             | a. Code written in machine language      b. three address code      c. Optimized machine code      d. Source program           |                      |           |
| <b>7</b>    | <b>Which data structure is commonly used for symbol table implementation?</b>                                                  |                      | <b>1</b>  |
|             | a. Queue      b. Stack      c. Hash Table      d. Linked List                                                                  |                      |           |
| <b>8</b>    | <b>Which parsing technique uses a stack to reduce input to the start symbol?</b>                                               |                      | <b>1</b>  |
|             | a. Recursive Descent Parsing      b. Top-Down Parsing      c. Bottom-Up Parsing      d. LL(1) Parsing                          |                      |           |
| <b>9</b>    | <b>In compiler which phases are machine dependent ?</b>                                                                        |                      | <b>1</b>  |

|             |                                                                                                                                                                                            |                                                       |                                                  |                                        |            |           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|----------------------------------------|------------|-----------|
|             | a. Lexical analysis and syntax analysis                                                                                                                                                    | b. Semantic analysis and intermediate code generation | c. part of Code optimization and code generation | d. Symbol table management             |            |           |
| <b>10</b>   | <b>What is a DAG used for in compilers?</b>                                                                                                                                                |                                                       |                                                  |                                        |            | <b>1</b>  |
|             | a. Syntax Analysis                                                                                                                                                                         | b. Semantic Analysis                                  | c. Code Optimization                             | d. Symbol Table Management             |            |           |
| <b>11</b>   | <b>Which phase of the compiler uses a regular expression?</b>                                                                                                                              |                                                       |                                                  |                                        |            | <b>1</b>  |
|             | a. Lexical Analysis                                                                                                                                                                        | b. Syntax Analysis                                    | c. Code Generation                               | d. Code Optimization                   |            |           |
| <b>12</b>   | <b>In an LR parser, what does "LR" stand for?</b>                                                                                                                                          |                                                       |                                                  |                                        |            | <b>1</b>  |
|             | a. Left-to-Right, Rightmost Derivation                                                                                                                                                     | b. Left-to-Right, Leftmost Derivation                 | c. Right-to-Left, Leftmost Derivation            | d. Right-to-Left, Rightmost Derivation |            |           |
| <b>Q. 2</b> | <b>Solve the following.</b>                                                                                                                                                                |                                                       |                                                  |                                        |            | <b>12</b> |
| <b>A)</b>   | <b>Write a note on YACC.</b>                                                                                                                                                               |                                                       |                                                  |                                        | <b>CO1</b> | <b>6</b>  |
| <b>B)</b>   | <b>Explain the importance of regular expression and context free grammar in compiler design.</b>                                                                                           |                                                       |                                                  |                                        | <b>CO1</b> | <b>6</b>  |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                                                                                                                |                                                       |                                                  |                                        |            | <b>12</b> |
| <b>A)</b>   | <b>Explain with example when we need to rewrite the grammar and why?</b>                                                                                                                   |                                                       |                                                  |                                        | <b>CO2</b> | <b>6</b>  |
| <b>B)</b>   | <b>Explain the use of symbol table in compiler with example.</b>                                                                                                                           |                                                       |                                                  |                                        | <b>CO2</b> | <b>6</b>  |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                     |                                                       |                                                  |                                        |            | <b>12</b> |
| <b>A)</b>   | <b>Give the algorithm to find FIRST and FOLLOW of grammar symbols and find FIRST and FOLLOW for the following grammar <math>G = \{ S \rightarrow AA, A \rightarrow a \mid b \}</math>.</b> |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>B)</b>   | <b>Construct the SLR parsing table for the following grammar <math>G = \{ S \rightarrow AA, A \rightarrow a \mid b \}</math>.</b>                                                          |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>C)</b>   | <b>Explain the SLR parsing table construction algorithm.</b>                                                                                                                               |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                     |                                                       |                                                  |                                        |            | <b>12</b> |
| <b>A)</b>   | <b>Explain Bottom-Up evaluation of inherited attributes.</b>                                                                                                                               |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>B)</b>   | <b>What is three address code? With one small example explain how a program is converted into three address code.</b>                                                                      |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>C)</b>   | <b>Explain the DAG and syntax tree with example. What is advantage of DAG.</b>                                                                                                             |                                                       |                                                  |                                        | <b>CO2</b> | <b>6</b>  |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                     |                                                       |                                                  |                                        |            | <b>12</b> |
| <b>A)</b>   | <b>Explain the issues in the design of code generation.</b>                                                                                                                                |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>B)</b>   | <b>Explain the principal sources of optimization.</b>                                                                                                                                      |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
| <b>C)</b>   | <b>Give the algorithm to construct the flow graph.</b>                                                                                                                                     |                                                       |                                                  |                                        | <b>CO3</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                         |                                                       |                                                  |                                        |            |           |